

# **AIAA 2290: Ethics, Privacy and Security in AI**

## **Sector-Specific Privacy Concerns and Innovations: Social Media**

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**Xuming HU**

**xuminghu@hkust-gz.edu.cn**

**The Hong Kong University of Science and Technology (Guangzhou)**

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1

Introduction to Social Media Privacy

2

Technology-Driven Privacy Threats

3

Cutting-edge Solutions

4

Future Scenarios



01

## Introduction to Social Media Privacy



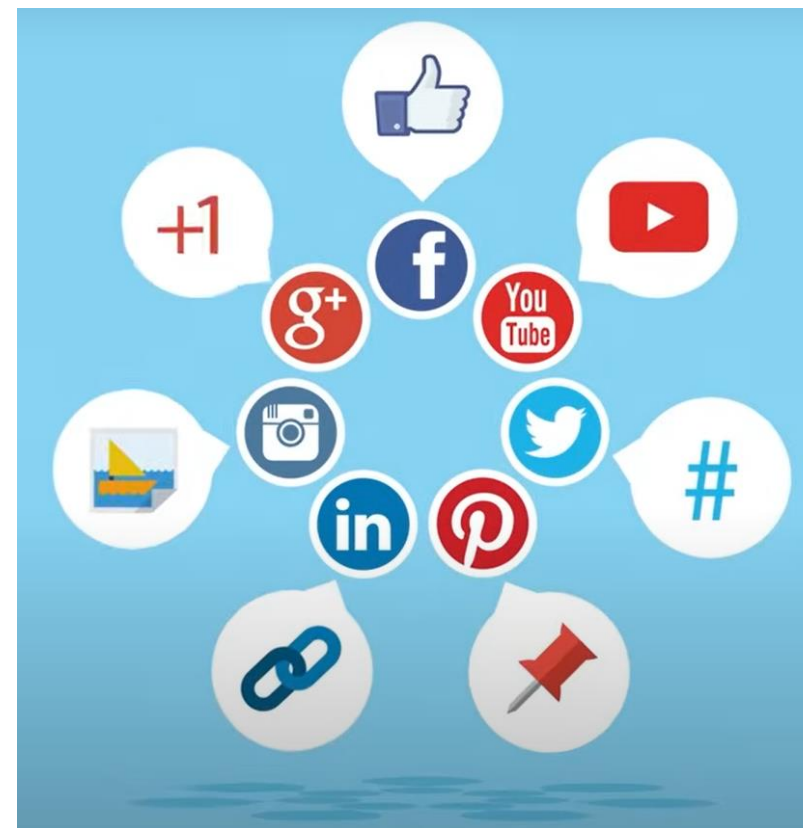
## Understanding Social Media Ecosystem



### Definition and Basic Concepts

Social media platforms are digital spaces **where users create, share, and interact with content.**

Privacy concerns arise due to the vast amount of personal data collected and utilized by these platforms.



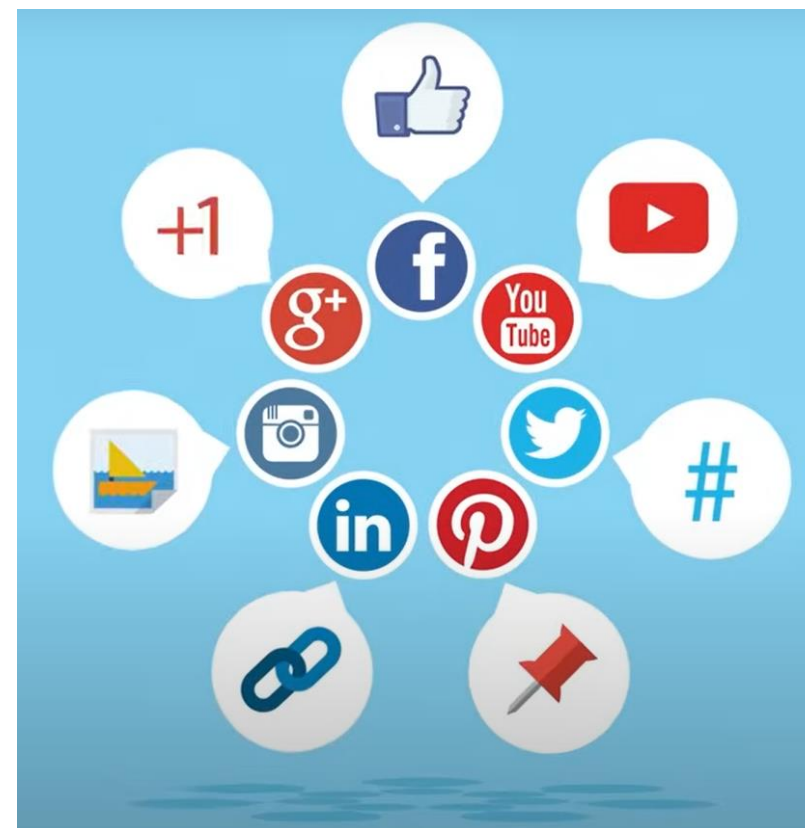


## Understanding Social Media Ecosystem



Privacy Fundamentals in Social Media

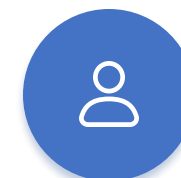
Privacy in social media revolves around **the protection of personal information, user consent, and the right to control how data is used.**





## Personal Information

When signing up for social media, users are typically required to provide personal information like name, email, birthdate, interests, location and more. All interactions and posts on social media are analyzed to understand user preferences and habits.





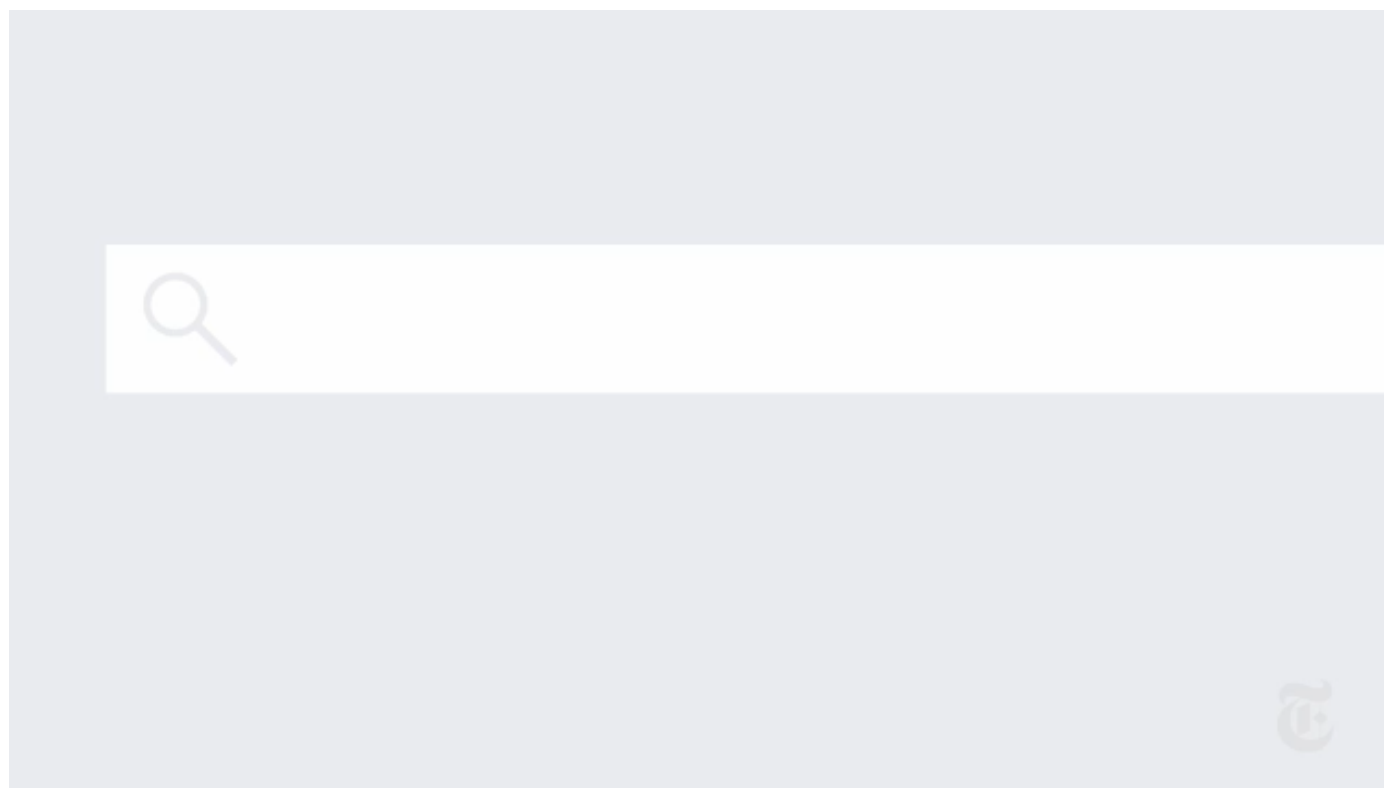
## Free Service → Data Monetization

# 01.

Business Model Explanation:

"Free Service → Data Monetization"

## How Facebook Tracks Your Data



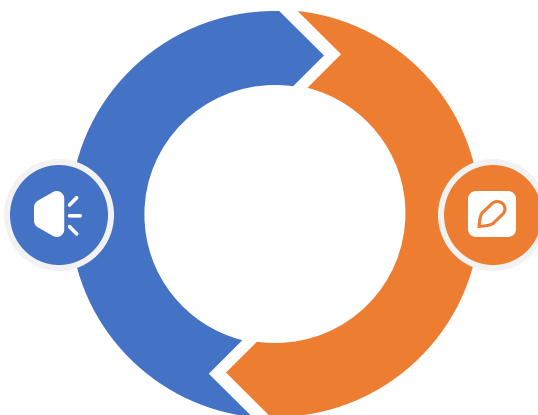




## Privacy Risks in Typical Scenarios

### Risks Associated with UGC Model

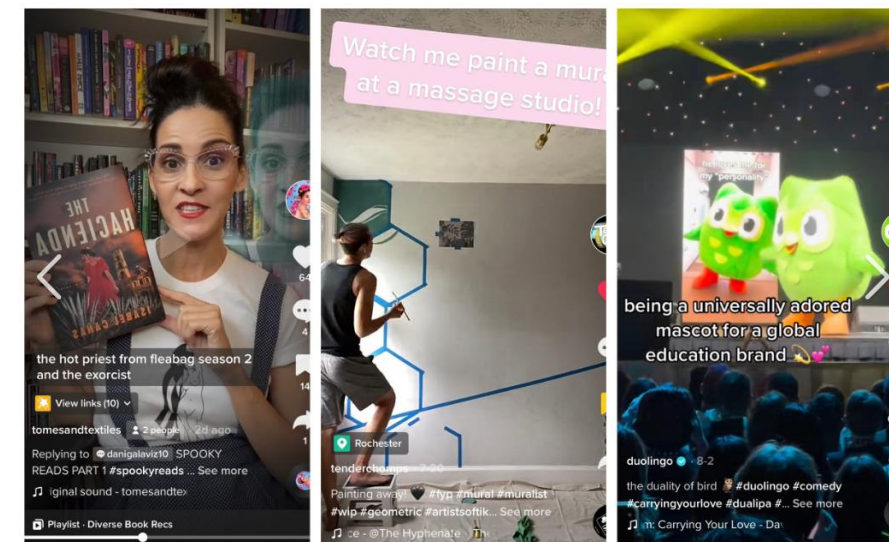
For instance, TikTok's video metadata can potentially expose sensitive user information if not properly secured.



What are the most-liked TikTok videos of 2021?



Here's the most liked TikTok videos of 2021. Picture: @thenickluciano via TikTok, @khaby.lame. via TikTok, @cacha.7.griselda via TikTok



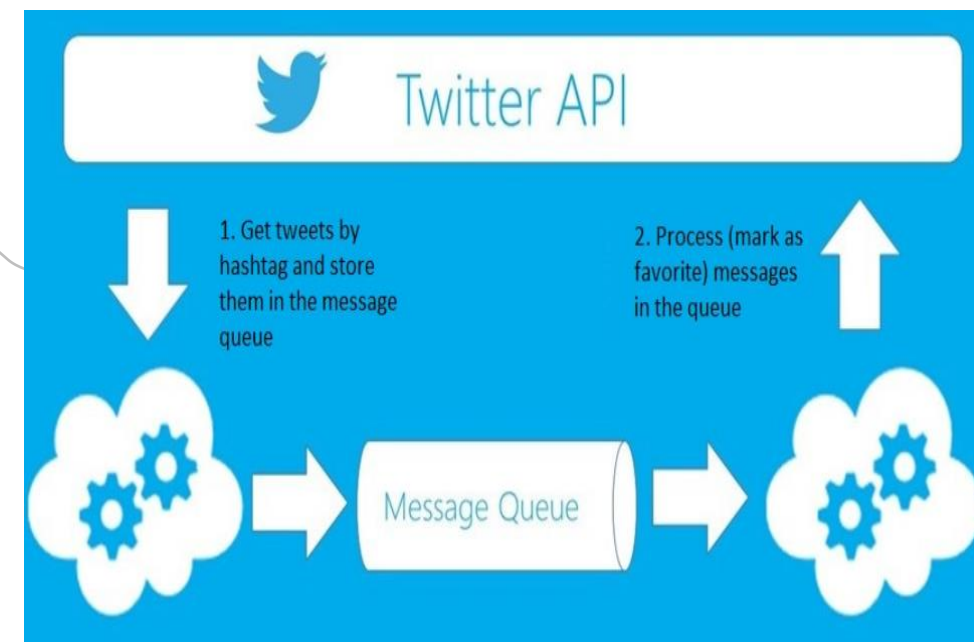
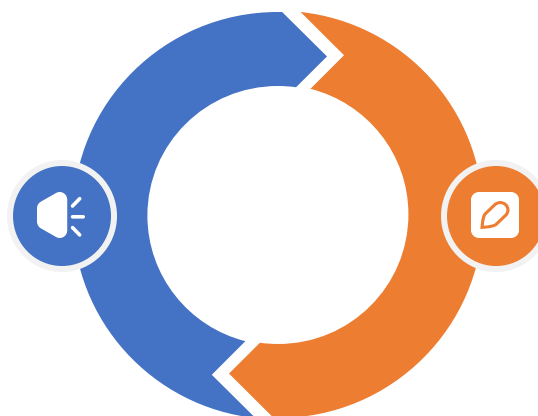




## Privacy Risks in Typical Scenarios

### Risks Posed by API Open Ecosystem

A case in point is the Twitter API, which can be exploited by malicious third- party developers, as illustrated in the attack surface analysis.





## Voting Question



Would You Pay for an Ad-Free Social Media Platform?



02

## Technology-Driven Privacy Threats



## What Are Cookies?

A cookie is a small piece of data from a specific website, that is stored on a user's computer while their browsing the web.





## Different functions

1. Keeping track of that user's browsing data in order to serve up targeted information such as ads for goods, or services.





## Different functions

2. Remembering your login details for a specific website.

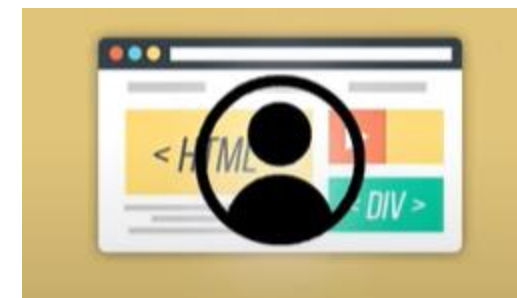
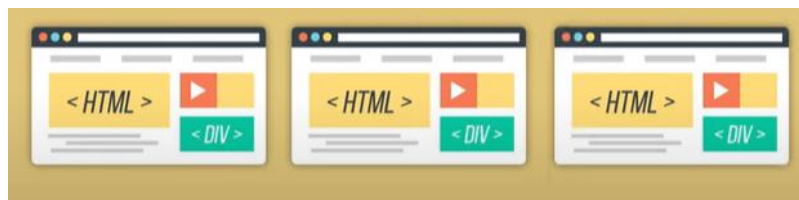






## Different functions

3. Cookies can also allow website owners to track exactly how many unique visitors they're getting to their website.

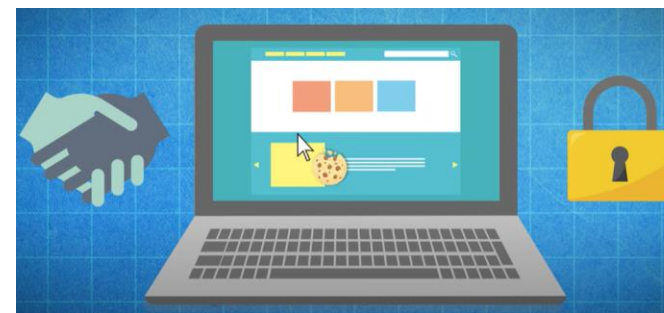






## Where did the cookie come from? And why?

The first cookie was invented in 1994 by a 24yr old programmer for NetScape Communications named Lou Montulli.





## How does a cookie work?

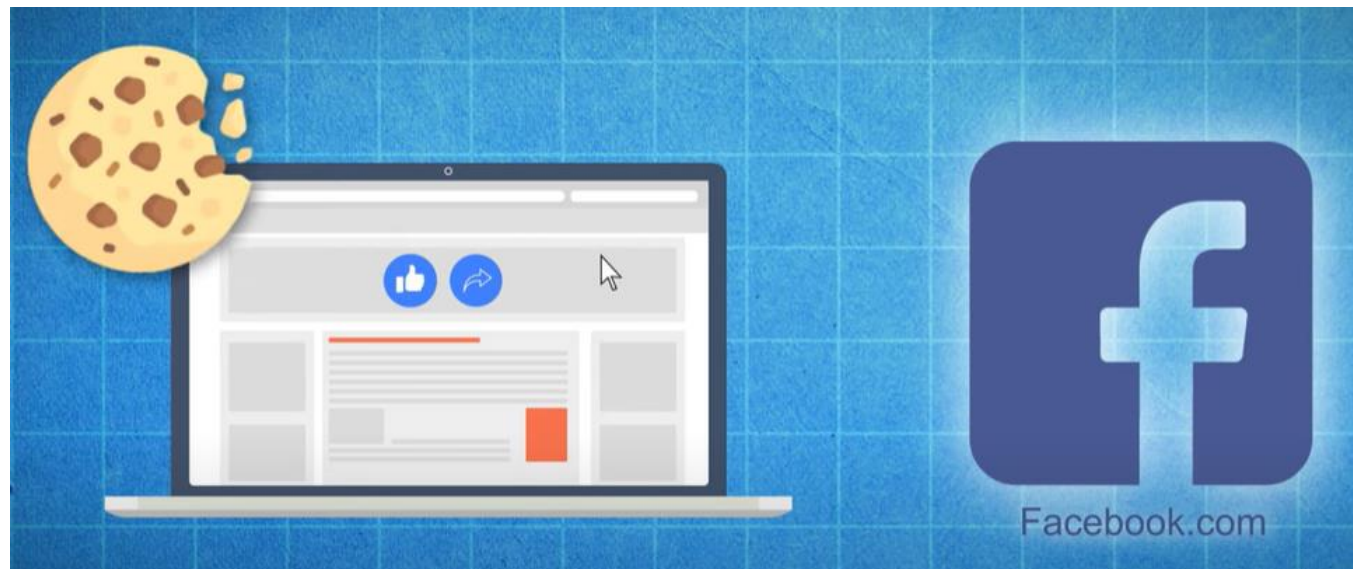
The site then uses this ID to keep track of your “session”.





## How does a cookie work?

There's another type of cookie called a third party cookie.





## *AI knowledge Quiz*

***What is a cookie in the context of web browsing?***

- A) A small piece of code that tracks hackers*
- B) A small piece of data stored on a user's computer by a website*
- C) A software program that encrypts user data*
- D) A type of cybersecurity malware*



## *AI knowledge Quiz*

***Which type of cookie is created by websites other than the one you are visiting?***

- A) First-party Cookies*
- B) Third-party Cookies*
- C) Session Cookies*
- D) Secure Cookies*



## The data hijacking scandal

Why you keep using Facebook, even if you hate it - YouTube

Facebook is under fire after admitting a political data firm improperly received access to profile data from tens of millions of users – and used it to try to sway voting habits.







## Facebook-Cambridge Analytica: A timeline

Facebook  
is desperately trying to mend  
its damaged reputation  
following the continued fallout  
of a massive data scandal.



Facebook CEO Mark Zuckerberg (C) leaves the office of Sen. Dianne Feinstein (D-CA) after meeting with Feinstein on Capitol Hill on April 9, 2018 in Washington, DC.

*Getty Images*





## Facebook-Cambridge Analytica: A timeline

April 21, 2010 — Facebook  
launches Open Graph





## Facebook-Cambridge Analytica: A timeline

2013 — 'This is your digital life'





## Facebook-Cambridge Analytica: A timeline

2014 — Rule changes

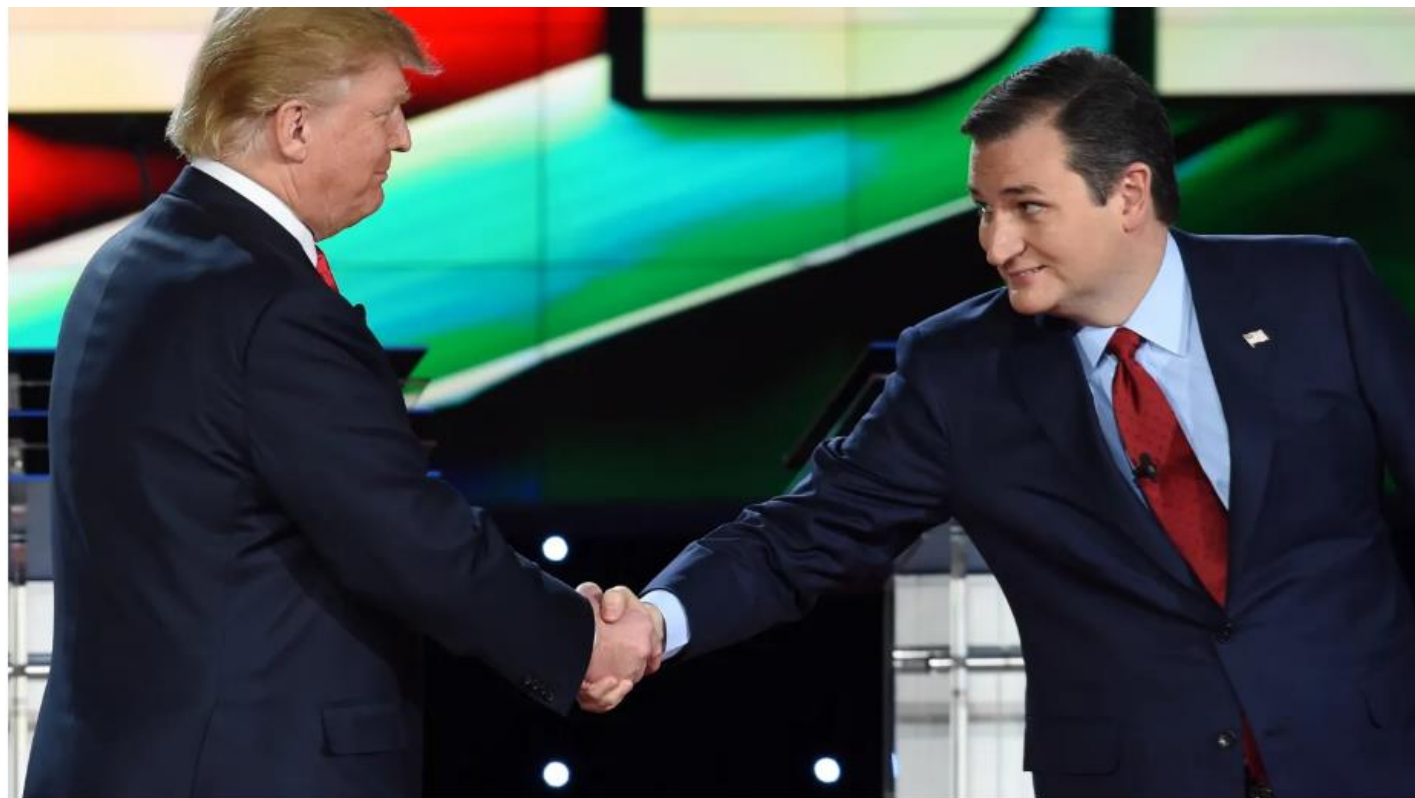






## Facebook-Cambridge Analytica: A timeline

December 11, 2015 – Ted Cruz



Republican presidential candidates businessman Donald Trump (L) and Texas Sen. Ted Cruz shake hands before the start of the Republican Presidential Debate in Las Vegas on Dec. 15, 2015.

*Robyn Beck / AFP / Getty Images*



## Facebook-Cambridge Analytica: A timeline

2016 — Donald Trump





## Facebook-Cambridge Analytica: A timeline

March 17, 2018 — Expose



Whistleblower Christopher Wylie on March 26, 2018, in London, England.

*Dan Kitwood | Getty Images*





## Facebook-Cambridge Analytica: A timeline

March 20, 2018 — FTC launch inquiry

March 21, 2018 — Zuckerberg breaks  
his silence



Mark Zuckerberg, chief executive officer and founder of Facebook

*Getty Images*



## Facebook-Cambridge Analytica: A timeline

March 25, 2018 — 'Sorry we didn't do more'

April 10, 2018 — Time to testify





03

## Cutting-edge Solutions



## Privacy Tech Innovation

01

**Snapchat** uses machine learning to power many features across the app - from sophisticated lenses, to serving relevant and interesting Discover content, to returning relevant search results, to security architecture that keeps our community safe.





## Device-Distributed Machine Learning

02

Device-Distributed Machine Learning (DDML) is a set of technologies that enable training of privacy-preserving machine learning models directly on client devices, without the need to transmit sensitive information to our servers.

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**Algorithm 1** DDML Algorithm for GLMs (client side). Parameters:

$Y$  - response value,  $\hat{Y}$  - predicted value.

$X$  - feature vector,  $B$  - a set of model weights.

$\gamma$  - learning rate,  $\epsilon$  - privacy budget.

$L(\mu, s)$  - Laplace distribution with mean  $\mu$  and scale  $s$ .

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Normalize response  $Y$  and features  $X$  to  $[0, 1]$

If  $N_c > 0$ , request model  $B^t$  from the server

Compute clipped average gradient:  $\|G\|_{[-1,1]} = \left\| \frac{1}{N_c} \sum_{i=1}^{N_c} (\hat{Y}_i - Y_i) X_i \right\|_{[-1,1]}$

Update model:  $B^{t+1} = B^t - \gamma \|G\|_{[-1,1]} + L\left(0, \frac{2\gamma}{\epsilon}\right)$

Return model  $B^{t+1}$  to the server

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**Algorithm 2** DDML Algorithm (server side). Parameters:

$k$  - the number of models,  $B$  - a set of model weights.

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Initialize  $k$  models

**for** each requested model update  $t$  **do**

    Pick a random model instance  $B^t$

    Send  $B^t$  to a client

    Receive updated  $B^{t+1}$  from a client

    Replace a random instance of the  $k$  models with  $B^{t+1}$

**end for**

Prediction: average  $k$  model instances

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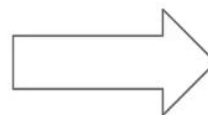
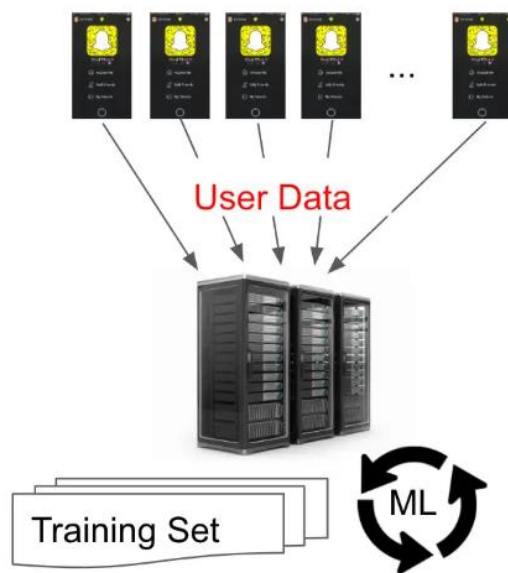




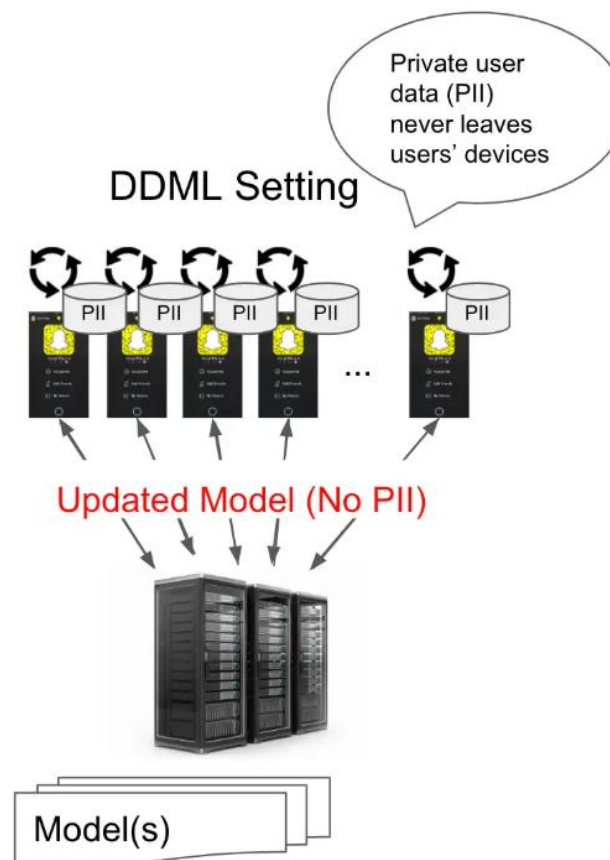
## Device-Distributed Machine Learning

03 The difference between the classical model training and DDML are shown schematically below.

Traditional Machine Learning Setting



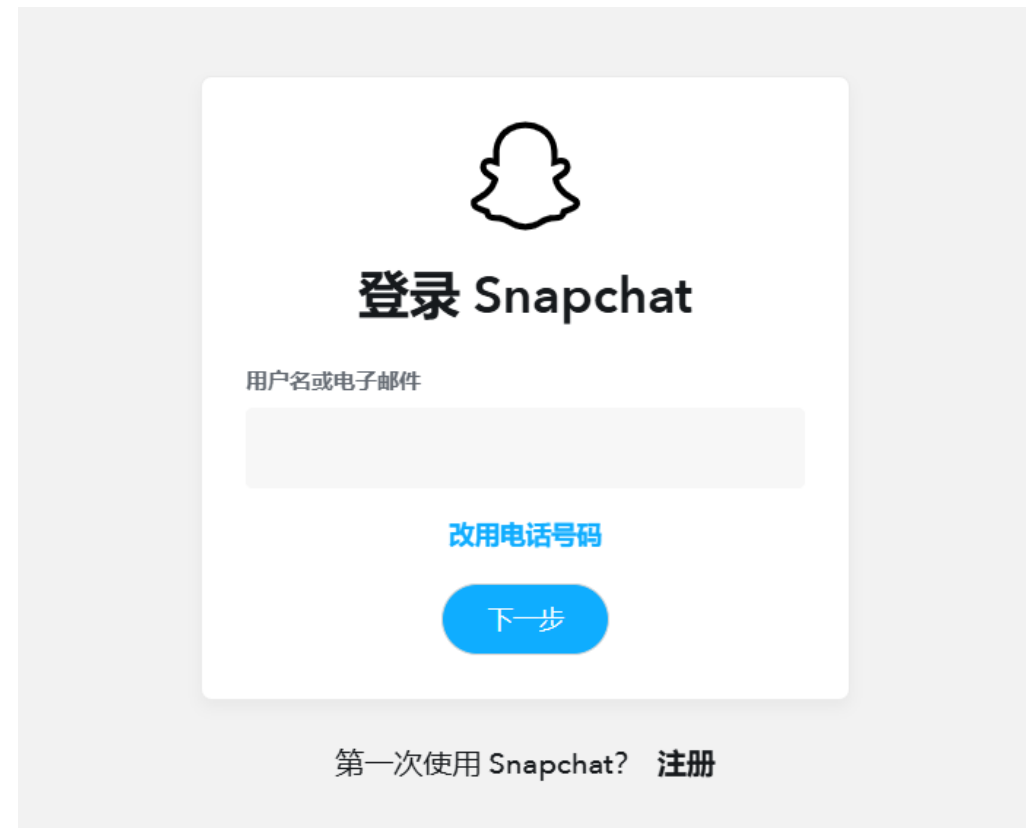
DDML Setting





### DDML Phishing Model

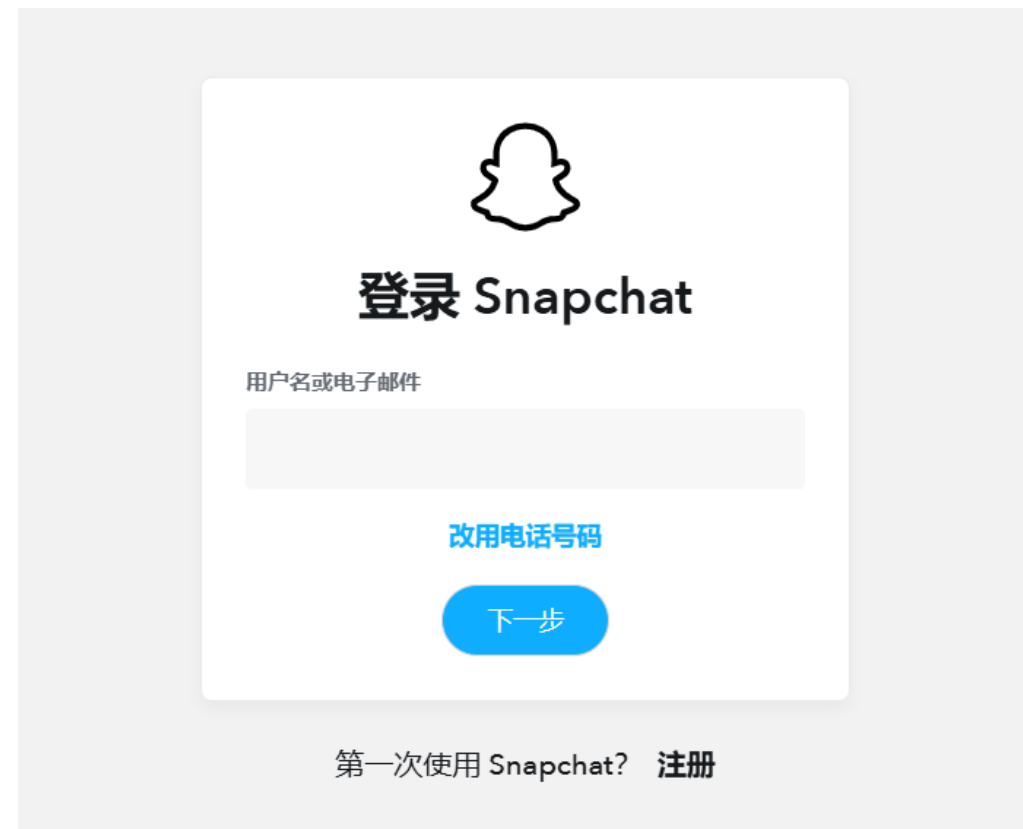
To give a practical example of DDML in action, it is useful to consider the problem of detecting phishy URLs. Stealing users' login and password information, commonly known as phishing, is an example of one-on-one abuse on the platform.





## DDML Phishing Model

the **URL** (its **length**, **number of query parameters**,  
**number of digits**, etc),  
**page** (does it have a **password box**, **number of paragraph**  
**tags**, etc)  
**context** (locale, time, country, etc)  
**prior knowledge** on whether this is a known phishy URL  
or not (**classification label**).





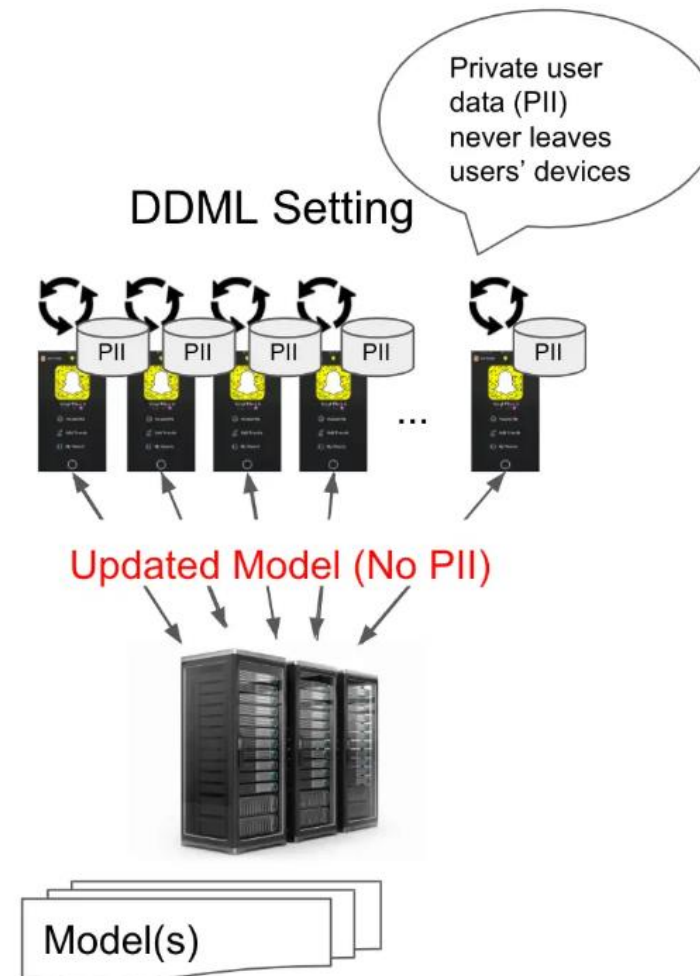
## Discussion

- **Advantages to DDML**

Privacy, scalability, resilience to failure and cost.

- **Disadvantages to DDML**

Need to ramp up on mobile development and understanding of iOS and Android platforms;  
Be mindful of device and network resources





## *AI knowledge Quiz*

***What is Device-Distributed Machine Learning (DDML)?***

- A) A method where all computations are done on a centralized cloud server*
- B) A framework where multiple devices collaboratively train a model without sharing raw data*
- C) A type of reinforcement learning used for gaming AI*
- D) A machine learning model that only runs on IoT sensors*





## *AI knowledge Quiz*

***Which of the following is an advantage of Device-Distributed Machine Learning?***

- A) Increased data privacy by keeping data on local devices*
- B) Requires less computational power on local devices*
- C) Eliminates the need for any network communication*
- D) Completely removes the need for AI model training*



## *AI knowledge Quiz*

***Which key technology enables Device-Distributed Machine Learning?***

- A) Federated Learning*
- B) Quantum Computing*
- C) Reinforcement Learning*
- D) Data Mining*



## Twitter working on abuse transparency report initiative





## Twitter working on abuse transparency report initiative



### USER REPORTS AND TOTAL ENFORCEMENT ACTIONS (EXCLUDING PLATFORM MANIPULATION & SPAM)

We strive to create an environment where users feel empowered to express themselves. When abusive behavior occurs, we make it easy for users to report violations of **X Rules** or local laws. Below, you'll find the total number of global user reports from January to June 2024, excluding Platform Manipulation and Spam, which is broken out separately below.

USER REPORTS 224,129,805

| POLICY AREA                                     | TOTAL      | % OF TOTAL REPORTS |
|-------------------------------------------------|------------|--------------------|
| Abuse & Harassment                              | 81,730,426 | 36.47%             |
| Child Safety                                    | 8,928,019  | 3.98%              |
| Hateful Conduct                                 | 66,896,539 | 29.85%             |
| Illegal or Certain Regulated Goods and Services | 35,695     | 0.02%              |
| Misleading & Deceptive Identities               | 5,179,431  | 2.31%              |
| Non-Consensual Nudity                           | 38,736     | 0.02%              |
| Private Content                                 | 9,902,566  | 4.42%              |
| Suicide & Self Harm                             | 2,479,097  | 1.11%              |
| Violent & Hateful Entities                      | 8,932,100  | 3.99%              |
| Violent Content                                 | 40,005,196 | 17.85%             |

### ENFORCEMENT ACTIONS

When we take **enforcement actions**, we may do so either on a specific piece of content (e.g., an individual post or Direct Message) or on an account. Below is a summary table of our automated and manual Rule Enforcement Actions and post violation rates that depict the percentage of all posts that were actioned for violating one of our platform rules during the covered time period. This data may include potential false positives and does not attempt to account for false negatives.

|                                |            |
|--------------------------------|------------|
| TOTAL ACCOUNT SUSPENSIONS      | 5,296,870  |
| TOTAL POSTS REMOVED OR LABELED | 10,675,980 |
| OVERALL POSTS VIOLATION RATE   | 0.0123%    |

| POLICY AREA                           | ACCOUNT SUSPENSIONS | POSTS REMOVED OR LABELED | POST VIOLATION RATE |
|---------------------------------------|---------------------|--------------------------|---------------------|
| Abuse & Harassment                    | 1,102,778           | 2,648,475                | 0.0030%             |
| Child Safety                          | 2,781,634           | 14,571                   | <0.0001%            |
| Hateful Conduct                       | 2,361               | 4,950,321                | 0.0057%             |
| Illegal or Regulated Goods & Services | 514,095             | 549,328                  | 0.0006%             |
| Misleading & Deceptive Identities     | 677,798             | 62                       | <0.0001%            |
| Non-Consensual Nudity                 | 52,093              | 156,498                  | 0.0002%             |
| Private Content                       | 1,719               | 34,497                   | <0.0001%            |
| Suicide & Self Harm                   | 2,286               | 86,184                   | 0.0001%             |
| Violent & Hateful Entities            | 57,185              | 153                      | <0.0001%            |
| Violent Content                       | 104,921             | 2,235,891                | 0.0026%             |



04

Future Scenarios



## Biometric Risks in Virtual Reality Social Platforms



### Horizon Worlds Eye Tracking Data Collection Alert

- Overview of VR social platforms (e.g., Horizon Worlds, Meta's VR ecosystem, Rec Room)
- Growing concerns over biometric data privacy
- Focus on eye-tracking data collection alerts





## What is Biometric Data in VR?



- Definition: Biometric data includes eye movements, facial expressions, hand gestures, etc.
- How VR platforms collect this data (e.g., built-in sensors, cameras)
- Why companies collect this data (e.g., user engagement, advertising, accessibility)



### Privacy and Security Risks



- **Data Sensitivity:** Eye movements reveal emotions, focus, and cognitive state.
- **User Profiling:** Companies can create detailed behavioral profiles.
- **Third-Party Access:** Risk of unauthorized data sharing.
- **Lack of Transparency:** Users are often unaware of data collection practices.



## How Web3 Decentralization Can Dismantle Big Tech Monopolies



2023 emerged as a breakout year for Web3 — a decentralized version of the internet built on open blockchain architectures — with the potential to disrupt the profit-hoarding data monopolies of Big Tech.

- Web3 empowers users through **blockchain** and **decentralization**, promising to transform data ownership models.



## Power Slowly Returns to Individual Users



- The path remains uphill, however, as blockchain-based models still need better UX and accessibility design to reach the non-tech mainstream.
- Meanwhile, Big Tech has also woken up to the threat and pours billions into its token and metaverse projects to co-opt decentralization trends.



## Conclusion

"Responsible Innovation Pyramid":

Technology Feasibility  
× Ethical Constraints  
× Public Interest

a formula involving technology feasibility, ethical constraints, and public interest.







### Propose New Privacy Measures for Social Media Platforms

- **Identify and address major privacy concerns** in social media platforms, such as user data tracking, third-party data sharing, and algorithmic profiling.
- Your proposed measures should **enhance user privacy** while ensuring usability and engagement.

#### Instructions:

**Step 1:** Identify key privacy risks in social media (e.g., targeted ads, facial recognition, misinformation tracking).

**Step 2:** Use AI tools like **ChatGPT or Deepseek** to brainstorm privacy-enhancing measures or analyze current social media privacy policies.

**Step 3:** Upload your proposal **on Canvas (10 minutes)** for peer review.

**Step 4: Bonus:** Those who actively share insights on how AI can improve social media privacy will earn a **small bonus reward**.

**Note:** This practice is **not included** in the final score but aims to develop critical thinking on social media privacy issues.



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Thank you

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